Version: 1.3



Agent Smith

Autonomous reasoning, code generation, and execution

Summary

In this project, you will build an **agentic framework** capable of autonomously solving coding challenges.
Your agent will reason, write code, execute it in a sandboxed environment, and iterate until a solution is found.
This project introduces **Code Agents**, **Model Context Protocol (MCP)**, and controlled code execution.
Made in collaboration with @ldevelle

#Agents #LLM #MCP



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Chapter 1

Foreword

Software engineering is no longer only about writing correct code, it is about **understanding systems, navigating large codebases, debugging failures, and iterating efficiently**. Modern LLMs generate code well, but fixing real bugs also needs exploration, hypothesis testing, execution, and observation.

This is where **Agent Smith** comes in. A Code Agent is not just a model that writes code: it **reasons about a task, interacts with tools, executes code safely, observes results, and adapts its strategy**. In this project, you will design and implement such a system, moving beyond static prompts and JSON-based tool calling to build a **fully agentic loop** driven by executable Python code.

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Chapter 2

AI Instructions

Context

During your learning journey, AI can assist with many different tasks. Take the time to explore the various capabilities of AI tools and how they can support your work. However, always approach them with caution and critically assess the results. Whether it's code, documentation, ideas, or technical explanations, you can never be completely sure that your question was well-formed or that the generated content is accurate. Your peers are a valuable resource to help you avoid mistakes and blind spots.

Main message

- Use AI to reduce repetitive or tedious tasks.
- Develop prompting skills — both coding and non-coding — that will benefit your future career.
- Learn how AI systems work to better anticipate and avoid common risks, biases, and ethical issues.
- Continue building both technical and power skills by working with your peers.
- Only use AI-generated content that you fully understand and can take responsibility for.

Learner rules:

- You should take the time to explore AI tools and understand how they work, so you can use them ethically and reduce potential biases.
- You should reflect on your problem before prompting — this helps you write clearer, more detailed, and more relevant prompts using accurate vocabulary.
- You should develop the habit of systematically checking, reviewing, questioning, and testing anything generated by AI.
- You should always seek peer review — don't rely solely on your own validation.

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Phase outcomes:

- Develop both general-purpose and domain-specific prompting skills.
- Boost your productivity with effective use of AI tools.
- Continue strengthening computational thinking, problem-solving, adaptability, and collaboration.

Comments and examples:

- You'll regularly encounter situations — exams, evaluations, and more — where you must demonstrate real understanding. Be prepared, keep building both your technical and interpersonal skills.
- Explaining your reasoning and debating with peers often reveals gaps in your understanding. Make peer learning a priority.
- AI tools often lack your specific context and tend to provide generic responses. Your peers, who share your environment, can offer more relevant and accurate insights.
- Where AI tends to generate the most likely answer, your peers can provide alternative perspectives and valuable nuance. Rely on them as a quality checkpoint.

✓ Good practice:

> I ask AI: "How do I test a sorting function?" It gives me a few ideas. I try them out and review the results with a peer. We refine the approach together.

✗ Bad practice:

> I ask AI to write a whole function, copy-paste it into my activity. During peer-evaluation, I can't explain what it does or why. I lose credibility — and I fail my activity.

✓ Good practice:

> I use AI to help design a parser. Then I walk through the logic with a peer. We catch two bugs and rewrite it together — better, cleaner, and fully understood.

✗ Bad practice:

> I let Copilot generate my code for a key part of my activity. It compiles, but I can't explain how it handles pipes. During the evaluation, I fail to justify and I fail my activity.

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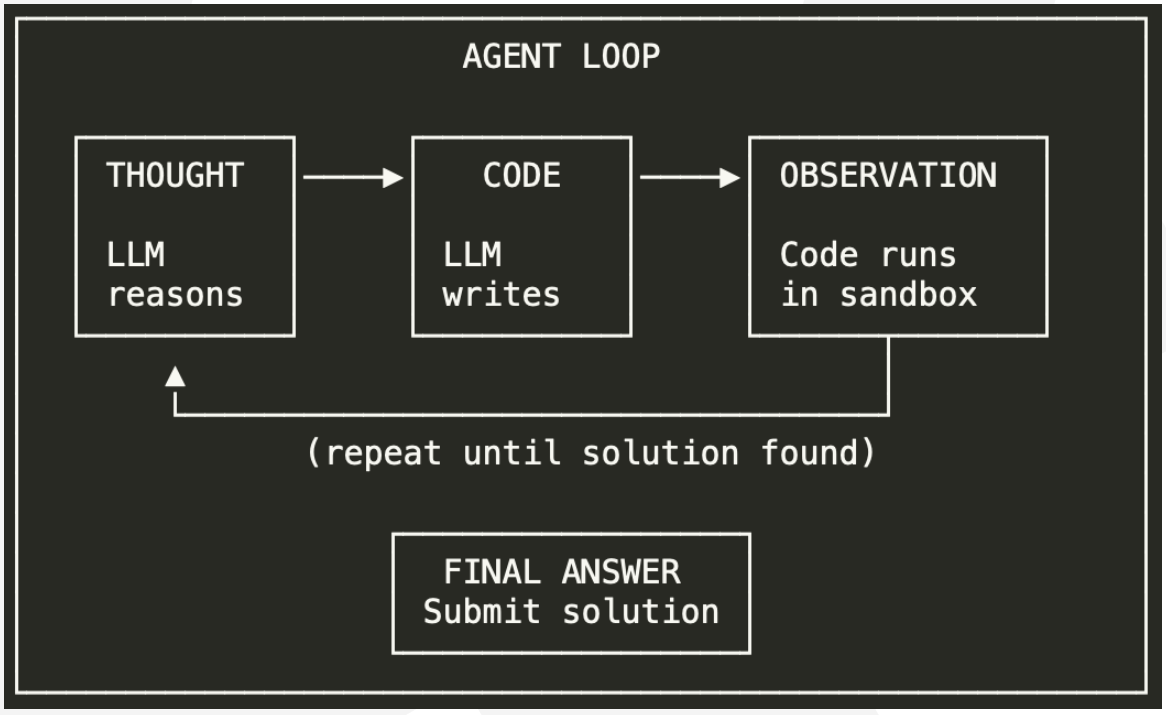
2.1 Overview

This project explores a new paradigm in AI-driven software engineering: **agentic code generation**.

Unlike classical approaches where an LLM outputs a final answer, your system will allow the model to:

- Write executable Python code
- Call tools directly from code
- Observe execution results
- Iterate until the task is solved

Your agent will operate through a structured loop: **Thought → Code → Observation**.



You will apply this framework to two different benchmarks:

- **MBPP**: algorithmic Python problems
- **SWE-bench**: real-world bug fixing in production repositories

The core challenge of this project is not only to make the agent intelligent, but also to make it **safe, controlled, reproducible, and measurable**.
In particular, you are expected to **evaluate and benchmark multiple language models** in order to analyze their performance and identify the most effective ones according to success rate and iteration efficiency.

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2.2 What is a Code Agent?

A **Code Agent** is an AI system that can:

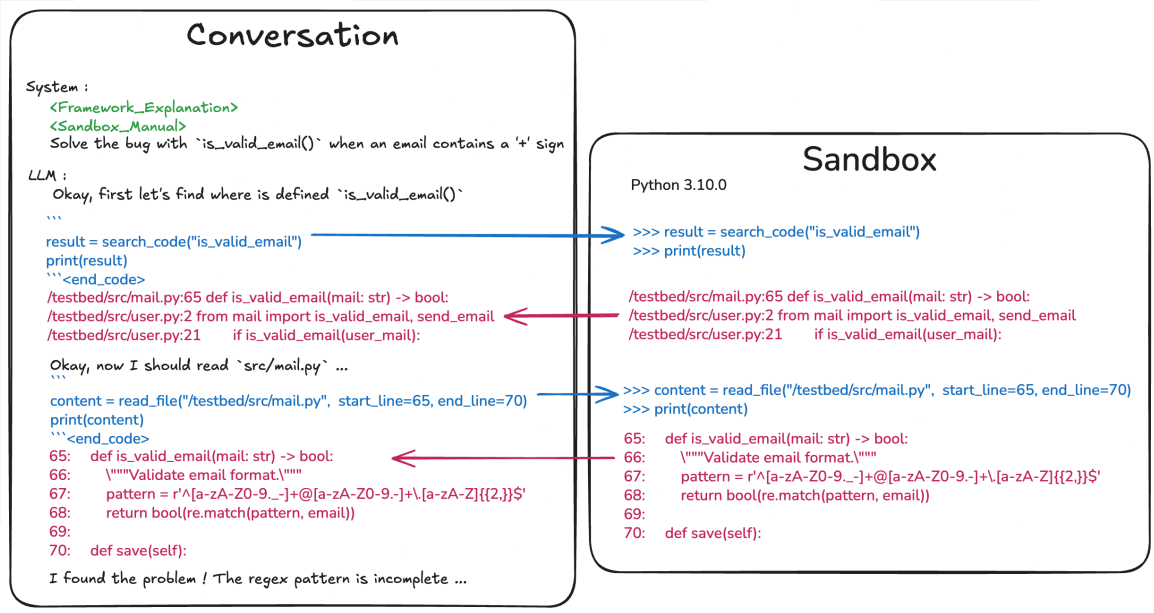
- Reason about a programming task
- Generate executable code
- Execute that code in a controlled environment
- Use tools to interact with files, tests, and repositories
- Observe the results and refine its approach

In this project, you will implement a **code-based tool calling** approach, where the LLM generates Python code such as:

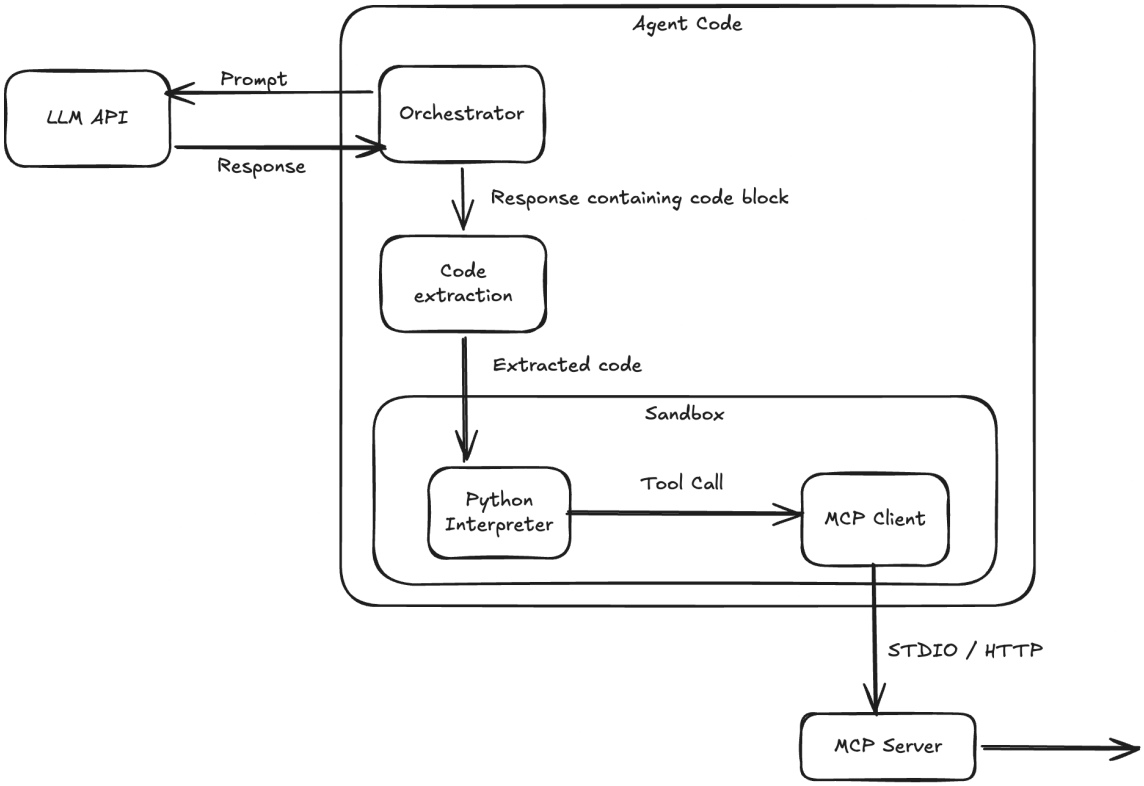
```
result = search_code("validate_email")
print (result)
content = read_file("models.py", 1, 50)
print (content)
```

This approach is more expressive than JSON tool calling, allowing:

- Persistent variables between steps
- Conditional logic and loops
- Complex multi-step reasoning



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1. **Agent/Orchestrator** is the central loop. It calls the LLM, extracts code, feeds it to the sandbox, reads observations, and repeats.

2. **Code Extraction** is a transform step between the LLM response and the sandbox.

3. **Sandbox** is the execution boundary. It enforces security restrictions on LLM-generated code. The sandbox contains an MCP client that connects to an external MCP server.

4. **final_answer()** is a built-in sandbox construct — it is NOT an MCP tool.

5. **MCP Server(s)** run as separate process(es) (via stdio or HTTP). Only your own MCP server(s) are allowed.

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Chapter 3

Common Instructions

3.1 General Rules

- You must use **Python 3.10**.
- You must use **uv** as your package manager.
- Your project must follow clean software architecture principles.
- All errors must be handled gracefully — crashes during evaluation will result in failure.
- Your code must be readable, structured, and documented.
- All execution must happen in a **sandboxed environment**.

3.2 Technical Constraints

- You must support multiple LLM providers and models.
- You must implement usage tracking (tokens, retries, latency, requests).
- You must implement a configurable sandbox (imports, filesystem access), as specified in Section 4.2.
- Your tools must work independently of the agent loop.
- You must not use any library that re-implements agent orchestration logic (e.g., llama-index, smolagents, langgraph, crewai, autogen).
- The agent loop must be your own implementation.
- Even though the project can be completed without it, multi-agent architectures are allowed, but the orchestration must be your own code.

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Chapter 4

Mandatory Part

4.1 Agentic Framework

In this project, you must build an **agentic framework** capable of autonomously solving coding challenges.

For LLM provider selection, see Section 4.6.1 for examples of free-tier APIs.

Your system must:

1. Implement a **Thought → Code → Observation** loop
2. Extract LLM-generated Python code from the model responses



When benchmarking across multiple providers (Section 4.7), your extraction layer should handle different output formats that LLMs produce — as well as to fit what the LLMs you use were trained on — not just Python code blocks:
A non-exhaustive list:

(a) **Python code blocks (primary)** — ```` ```python ... ``` <end_code> ````

(b) **XML tool calls** — Anthropic-style `<invoke name="..."><parameter>...</parameter></invoke>`

(c) **JSON/Hermes tool calls** — `<tool_call>{"name": "...", "arguments": {...}}</tool_call>`

(d) **ReAct format** — `Action: tool_name` / `Action Input: {...}`

Non-Python formats should be converted to equivalent Python function calls (e.g., `result = read_file(filepath="/testbed/file.py")`) before sandbox execution. This lets the sandbox remain format-agnostic while supporting any model.

3. Execute the generated code inside a sandboxed environment

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4. Feed the sandbox execution results back to the LLM
5. Solve benchmark tasks autonomously using the agent loop
6. Design the **system prompt**, including:
    - clear documentation of the available tools
    - examples of structured response slots (e.g. `Thought`, `Code`, `Observation`)
    - examples of effective agent reasoning loops



Your sandbox must provide explicit feedback to the LLM in all of these situations:

- No valid code block was found in the model's response
- A code block was malformed but was interpreted anyway (explain how)
- Execution hit the timeout and output is partial
- Tool output was truncated due to size limits
- An edit introduced a syntax error or lint violation

The LLM should never be left guessing about what happened. Silent failures lead to hallucinated observations and wasted iterations.

4.1.1 Common Output Models

Both MBPP and SWE-bench agents produce the same `StepMetrics`/`SolutionOutput` shape; only the `benchmark` and `solution` field semantics change (Python code for MBPP, a git patch for SWE-bench).

```
class StepMetrics(BaseModel):
    """Metrics for a single agent step.

    Each step corresponds to one LLM generate -> sandbox execute cycle.
    All fields are required for evaluation, empty strings are acceptable
    for steps where a field doesn't apply (e.g., no sandbox execution).
    """
    step: int = Field(..., description="1-indexed iteration number")
    input_tokens: int = Field(..., description="Tokens sent to the LLM for this step
        ")
    output_tokens: int = Field(..., description="Tokens generated by the LLM for
        this step")
    request_time_ms: float = Field(..., description="Wall-clock time for the LLM API
         call in milliseconds")
    timestamp: str = Field(default_factory=lambda: datetime.now().isoformat(),
        description="ISO 8601 timestamp of when this step was recorded")
```

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```
    api_url: str = Field(default="", description="Base URL of the LLM API endpoint (
        e.g., 'https://openrouter.ai/api/v1')")
    model_name: str = Field(default="", description="Model identifier used for this
        step (e.g., 'qwen/qwen3-235b-a22b-2507')")
    llm_output: str = Field(default="", description="Raw text generated by the LLM
        before code extraction")
    sandbox_input: str = Field(default="", description="Python code sent to the
        sandbox for execution")
    sandbox_output: str = Field(default="", description="Sandbox execution result (
        stdout/stderr/error message)")
    retries: int = Field(default=0, description="Number of LLM API retries before
        getting a successful response (0 = first attempt succeeded)")


class SolutionOutput(BaseModel):
    """Output from student solution, required format for evaluation.

    This is the JSON structure your agent must produce and write to solution.json.
    The moulinette validates this against task correctness and metrics limits.
    """
    task_id: str = Field(..., description="Task identifier (MBPP task_id as string,
        or SWE-bench instance_id)")
    benchmark: str = Field(..., description="Benchmark type: 'mbpp' or 'swebench'")
    success: bool = Field(..., description="Whether the agent believes it solved the
         task")
    solution: str = Field(..., description="For MBPP: the Python function code. For
        SWE-bench: the git patch (diff)")
    iterations: int = Field(..., description="Number of agent loop iterations used")
    total_requests: int = Field(..., description="Total number of LLM API requests
        made (including retries)")
    total_input_tokens: int = Field(..., description="Sum of input_tokens across all
         steps")
    total_output_tokens: int = Field(..., description="Sum of output_tokens across
        all steps")
    total_time_seconds: float = Field(..., description="Wall-clock time from agent
        start to finish")
    steps: List[StepMetrics] = Field(default_factory=list, description="Per-step
        metrics, one entry per agent iteration")
    system_prompt: str = Field(default="", description="Full system prompt sent to
        the LLM (for provenance checking)")
    error: Optional[str] = Field(default=None, description="Error message if the
        agent failed (None if successful)")
    timestamp: str = Field(default_factory=lambda: datetime.now().isoformat(),
        description="ISO 8601 timestamp of when the solution was produced")
```



Each step in the steps array of SolutionOutput must include api_url, model_name, fields identifying the LLM endpoint used for that step, plus llm_output (raw LLM response), sandbox_input (code sent to sandbox), sandbox_output (execution result), and retries (API retry count).

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4.1.2 Development Approach

Building an autonomous agent is a complex task. Here are questions to guide your development:

Starting out: Which benchmark is simpler and should be tackled first? Am I testing with the most capable model available, or handicapping myself with constraints too early? Can my system solve the easiest task with token and iteration limits temporarily removed? If not, adding constraints will only make it worse.

Debugging your agent: What happens in the first 3–5 iterations: tool-calling struggles, hallucinated observations, or an unexpected path? Have I tried solving the target task myself, manually, and does my prompt reflect that methodology? For SWE-bench, break the eval script into smaller goals and track whether individual tests pass rather than waiting for the full script to succeed.

Scaling up: Does a solution that works on one task generalize, or did I overfit? Am I optimizing too early (tokens, model choice), or have I proven the approach works first? Different models prefer different tool-calling styles: am I fighting the model's natural tendencies, or working with them?

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4.2 The Sandbox

This part focuses on the design and implementation of a secure and configurable execution sandbox for the agent.

1. Implement a **sandbox CLI usage**

```
# Launch interactive sandbox
uv run sandbox

# With custom configuration
uv run sandbox sandbox_template.json

# With MBPP tools (stdio)
uv run sandbox --mcp-stdio "python mcp_tools_mbpp.py" sandbox_template.json

# With MBPP tools (HTTP)
uv run sandbox --mcp-server <URL>

# With SWE-bench tools
uv run sandbox --mcp-stdio "python mcp_tools_swebench.py" sandbox_template.json
```

The **interactive sandbox** (`uv run sandbox` with no task argument) is a REPL-style command-line mode. It must open a prompt, read user-typed code in a loop, and execute each entry inside the sandbox namespace, subject to the same import, filesystem, timeout and memory restrictions defined in Section 4.2, with the connected MCP tool wrappers and `final_answer` available. After each entry it prints the result or any raised error and returns to the prompt. It exits cleanly on the `exit` command or on EOF (`Ctrl+D`).

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2. Implement a `final_answer` tool

   What `final_answer` IS:

   `final_answer` is a callable function that the sandbox injects into the execution namespace. When the agent's code calls `final_answer(answer_string)`, the sandbox captures the argument and signals to the agent loop that the task is complete. The agent loop then terminates and produces the `SolutionOutput`.

   What `final_answer` is NOT:

   `final_answer` is **not** an MCP tool — it is not provided by any MCP server. It is always available in the sandbox namespace regardless of which MCP server is connected. MCP tools operate outside the sandbox (e.g., reading files in Docker, running tests), while `final_answer` operates within the sandbox to control the agent loop.

   How to use `final_answer`:

   - **MBPP**: call `final_answer(your_solution_code)` — pass your Python code as the argument
   - **SWE-bench**: call `final_answer(get_patch())` — pass the git patch retrieved via `get_patch()` as the argument

   Architectural boundary:

   The sandbox provides two kinds of callable functions in the execution namespace:

   (a) **MCP tool wrappers** — discovered dynamically from the connected MCP server

   (b) `final_answer` — always present, provided by the sandbox itself

   When you connect a different MCP server, the MCP tool wrappers change but `final_answer` remains.

   Exception propagation:

   Your sandbox must correctly propagate exceptions that control program flow. In particular, `KeyboardInterrupt` and `SystemExit` must not be silently caught — they need to reach the agent loop for proper shutdown.

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3. Enforce sandbox security constraints: Running LLM-generated code is inherently risky. Your sandbox is not just a technical component — it is a **safety boundary** between an autonomous system and the real world.

    - **Import restrictions**: Only modules from the configured allowlist may be imported
    - **Filesystem restrictions**: file access by the sandboxed code is limited to an allowlist of directories (the `allowed_directories` field of `SandboxConfig`). These are paths on the filesystem the sandbox process itself sees, evaluated inside the sandbox, not host-only paths: any directory the code must reach has to be in the allowlist. Several entries are allowed so you can keep the task workspace (e.g., `/testbed`) separate from a writable scratch/runtime area (e.g., `/tmp/agent`); anything outside the allowlist is denied.
    - **No network access**: Prevent any outbound or inbound network connections
    - **Execution timeout**: Terminate code exceeding the configured timeout (applies only to sandboxed code)
    - **Memory limits**: Terminate code exceeding allowed RAM usage
    - **Restricted builtins**: Remove or override dangerous builtins to prevent privilege escalation

4. Integrate an **MCP server**

    - Mandatory tools must be integrated into the agent
    - MCP tools, resources, and prompts must be exposed
    - MCP tools must be callable as Python functions from the sandbox
    - The system will be tested with an **unknown MCP server**
    - The MCP tool files (mcp_tools_mbpp.py, mcp_tools_swebench.py) should be located at the root of your repository
    - Both stdio and streamable HTTP transports must be supported for MCP server connections

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5. Generate a sandbox manual to be fed to the LLM prompt, which must include the MCP tools doc, or how to access it.



The sandbox manual should be dynamically generated from the connected MCP server's tool schemas — tool names, descriptions, and parameter types. When a different MCP server is connected, the manual should automatically reflect that server's tools.
The manual is what the LLM reads to understand what tools are available and how to call them.

Sandbox limits and behavior must be configurable using **Pydantic models and JSON configuration files**.

```
class SandboxConfig(BaseModel):
    """Sandbox configuration for student solutions.

    Uses allowlist approach: only imports in authorized_imports are allowed.
    Everything else is blocked by default.
    """
    authorized_imports: List[str] = Field(default_factory=lambda: [
        "math", "math.*",
        "collections", "collections.*",
        "itertools", "re", "json",
        "typing", "typing.*",
        "functools", "operator",
        "heapq", "bisect", "copy",
        "string", "random",
        "datetime", "datetime.*",
        "array", "cmath",
    ])
    allowed_directories: List[str] = Field(default_factory=lambda: [
        "/testbed", "/tmp/agent"
    ])
    max_execution_time_seconds: int = 30
    max_memory_mb: int = 512
```

An entry ending in `.*` also allows importing that module's submodules (e.g. `collections.*` permits `collections.abc`).

When a different MCP server is connected, the sandbox should dynamically discover and expose that server's tools. Your mandatory tools are only present when your own MCP server is connected.

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The sandbox is the central execution layer. It connects to an MCP server and exposes its tools as callable Python functions within the sandbox namespace. The sandbox wraps the MCP client, not the other way around.
The sandbox and MCP tools are independent security domains: the sandbox restricts what LLM-generated Python code can do (imports, paths, timeout, memory), while MCP tool actions happen outside the sandbox and are not subject to the sandbox timeout (e.g. a tool spawning an external process).

You are expected to implement the security mechanisms using only standard Python libraries and builtins. No external packages like RestrictedPython are allowed.

Sandbox isolation approaches



Think about it: untrusted code inside your own process or in a separate one? Each choice trades off security boundaries, timeout handling (how do you kill runaway code?), and communication (how do you get results back?). Multiple valid approaches exist, pick what fits your architecture.

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4.3 MBPP Agent

In this part, you will implement an autonomous agent dedicated to solving **MBPP (Mostly Basic Python Problems)** tasks.

1. Implement an **agent CLI interface**

```
# 1. Dump a task
cd moulinette
uv run moulinette_eval dump mbpp --output ../cache/mbpp_task.json

# 2. Run your agent
cd ../student
uv run python -m agent_mbpp --task-file ../cache/mbpp_task.json \
--output ../cache/mbpp_solution.json \
--model-name "model/name" --provider-url "https://provider.api/v1"

# 3. Validate solution
cd ../moulinette
uv run moulinette_eval validate mbpp ../cache/mbpp_task.json \
../cache/mbpp_solution.json
```

   - Task loading
   - Agent execution

2. Implement **MBPP MCP tools**
   - `run_tests(code, test_list)`
     Run a candidate solution against the given test assertions. It returns a JSON string with a `success` boolean (whether all assertions passed) and an `output` field.
   - You may implement any additional tools you consider useful

3. Define **Pydantic models** for:
   - Task input

```
    class MBPPTaskInput(BaseModel):
"""Input for MBPP task evaluation."""
task_id: int
task_definition: str
function_definition: str
test_imports: List[str] = Field(default_factory=list)
test_list: List[str] = Field(default_factory=list)
```

   - Agent output: same `StepMetrics`/`SolutionOutput` models as in Section 4.1 (`benchmark="mbp`
     `solution` = your function code)

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4. Design your agent to operate within the limits defined in Section 5.1 (iterations, tokens, time). The evaluation validates that your agent's output stays within these limits. `max_iterations` should be a configurable parameter of your agent loop.

External LLM providers are allowed, provided the interface remains compliant with the project constraints.

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4.4 SWE-bench Agent

This part focuses on implementing an autonomous agent capable of solving **SWE-bench** tasks inside Dockerized environments.

For SWE-bench, both approaches are valid:
(a) deploy the sandbox inside the Docker container, or
(b) run the sandbox on the host with MCP tools bridging into Docker.
Regardless of choice, sandbox security constraints must be enforced.

- Fix real bugs or implement features in real repositories
- Explore codebases inside Docker containers, you are responsible to clean it after your program execution
- Generate and submit valid patches using 'git -c core.fileMode=false diff'



When testing your SWE-bench tools in isolation, the moulinette sets the environment variable `TESTBED_PATH` to the repository root before starting your MCP server. Your tools must read this exact variable name to locate the repository.

You may install additional dependencies inside the Docker container (e.g., ruff, jedi, tree) to enhance your tools.



For initial testing, try the simplest task first: sympy__sympy-14711, sympy__sympy-13480, pydata__xarray-4629. What happens when you remove unused tools from your setup?

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1. Implement an **agent CLI interface**

```
# 1. Dump a task
cd moulinette
uv run moulinette_eval dump swebench --output ../cache/swebench_task.json

# 2. Run your agent
cd ../student
uv run python -m agent_swebench --task-file ../cache/swebench_task.json \
--output ../cache/swebench_solution.json \
--model-name "model/name" --provider-url "https://provider.api/v1"

# 3. Validate solution
cd ../moulinette
uv run moulinette_eval validate swebench ../cache/swebench_task.json \
../cache/swebench_solution.json
```

2. Implement **SWE-bench MCP tools**

   - Tools must follow the specification defined in Mandatory Tools
   - You may implement any additional tools you consider useful

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3. Define **Pydantic models** for:

   - Task input

```
class SWEBenchTaskInput(BaseModel):
    """Input for a SWE-bench task, provided by the moulinette.

    Your agent receives this and must produce a git patch that fixes the
        issue.
    """
    instance_id: str = Field(..., description="SWE-bench instance
        identifier (e.g., 'sympy__sympy-23534')")
    problem_statement: str = Field(..., description="The GitHub issue
        description, what needs to be fixed")
    docker_image: str = Field(..., description="Full Docker image name to
        pull (e.g., 'swebench/sweb.eval.x86_64.sympy_1776_sympy-23534:
        latest')")
    eval_script: str = Field(..., description="Bash script to run inside
        the container to evaluate the patch")
    hints_text: str = Field(default="", description="Optional hints about
        the issue (may be empty)")
    repo: str = Field(default="", description="Repository name (e.g., '
        sympy/sympy')")
```

   - Agent output (patch): same `StepMetrics`/`SolutionOutput` models as in Section 4.1 (`benchmark="swebench"`, `solution` = your git patch)

4. Enforce **hard limits** as defined in Section 5.1

External LLM providers are allowed, provided the agent remains fully compatible with the evaluation infrastructure.

`SolutionOutput` must include a `system_prompt` field containing the full system prompt sent to the LLM. Each step in `steps` must include `llm_output`, `sandbox_input`, and `sandbox_output` fields logging the raw LLM response, the Python code sent to the sandbox, and the execution result. A `retries` field tracks how many LLM API retries were needed for the step.

```
{
  "steps": [
    {
      "step": 1,
      "llm_output": "I'll read the file to understand the module.\n```python\nresult
          = read_file(filepath=\"/testbed/src/module.py\")\n```",
      "sandbox_input": "result = read_file(filepath=\"/testbed/src/module.py\")",
      "sandbox_output": "def solve(x):\n return x + 1\n...",
      "retries": 0,
      "input_tokens": 1234,
      "output_tokens": 567,
      "..."
    }
```

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```
  ]
}
```

This metadata is inspected during evaluation to verify the agent solved tasks through legitimate code exploration and reasoning, not by looking up solutions from pull requests, issues, or external sources.

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4.5 Mandatory Tools

You must implement all the following tools. Each tool is exposed by the MCP (Model Context Protocol) server and will be tested independently in the context of the SWE-bench benchmark.

4.5.1 File System Tools

- `read_file(filepath, start_line, end_line)`
  Read the content of a file with line numbers.
  The output format must be similar to `cat -n`:

```
<line_number>: <line_content>
<line_number>: <line_content>
...
```

- `edit_file(filepath, old_str, new_str)`
  Replace an exact string in a file with a new string.

- `list_files(directory, pattern)`
  List files in a directory matching a given pattern.

4.5.2 Code Search Tools

- `search_code(pattern, file_pattern)`
  Perform a grep-like search in the codebase.
  The output must follow this format:

```
/absolute/path_to_file.py:<line_number> <line_content>
/absolute/path_to_other_file.py:<line_number> <line_content>
...
```

- `search_function_or_class_definition_in_code(name)`
  Find the definition of a function or a class.
  The output format must be similar to search_code format.

- `find_references(name, filepath, line)`
  Find all usages of a symbol (function or class).
  The output format must be similar to search_code format.

4.5.3 Execution Tools

- `run_tests()`
  Execute the evaluation script.

- `get_patch()`
  Retrieve the unified `git diff` of all changes made to the repository, depending on the implementation.

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- `run_command(command, workdir)`
  Execute a shell command in the specified working directory.
  Returns the command's stdout, stderr, and exit code.

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4.6 LLM API Providers

To implement your agent, you will need to perform calls to **Large Language Model (LLM) APIs**. Several providers offer free tiers or usage quotas that are sufficient for development and experimentation.

The list below is provided for **illustrative purposes only**. It is based on publicly available resources and may evolve over time.
You are free to use other providers as long as your system complies with the project requirements (free tier, multiple API tokens per provider).

The dataset your agent is evaluated on is called SWE-bench Verified. Knowing this, you can find:

- **Model leaderboards** — which LLMs perform best on real-world coding tasks?
- **Agent system descriptions** — how do the top-performing systems design their agent loop, tools, and prompts?
- **Per-task traces and evaluations** — how does a specific model perform on the exact task you're iterating on? This can help you choose the right model for your first debugging cycles.



What happens if the model keeps generating after a tool call instead of waiting for real execution output? Use a `stop_sequences` (or `stop`) API parameter (e.g. `<end_code>`, `</tool_call>`) to stop generation at the token that ends a code block, otherwise the model may hallucinate fictional tool output instead of waiting for the real one.

4.6.1 Examples of Free Providers

Examples include OpenRouter, Together AI, Groq, Google AI Studio (Gemini), Mistral AI, Cohere, Fireworks AI, Perplexity AI, and Anyscale (non-exhaustive).

Ensure your implementation supports **multiple API keys per provider** and consider implementing provider fallback. Rate limits and quotas may vary, especially under load during evaluation.
Important:

- This list is **not exhaustive and not contractual**.
- Access conditions, quotas, and available models may change over time.
- Learners are encouraged to explore additional providers or to use self-hosted models if needed.
- **The entire solution must rely exclusively on free tiers.**
  - No paid plans, purchased credits, or billing-enabled accounts are allowed.
  - The project must be fully executable using only free quotas at evaluation time.
- **Multi-token management is mandatory.**

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- Your system must support **multiple API tokens per provider**.
- Token rotation must be implemented to handle rate limits and quota exhaustion.

- Your implementation must be sufficiently **abstract** to allow switching providers without major refactoring.



The choice of provider is not graded.
The quality of your abstraction, error handling, and overall architecture is.

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4.7 Model Benchmark Report

Raw solve rate only tells part of the story of which model is best for your agent, exploration efficiency, token cost, provider reliability, and iteration discipline matter too. You must produce a benchmark report (`BENCHMARK_REPORT.md` at the root of your repository) comparing at least **5 models** across at least **2 providers** on the same set of at least **3 SWE-bench tasks**.



Free-tier providers enforce daily token/request quotas, and running 5 models can exhaust one before you're done. Since more providers means more independent quotas to draw from, we recommend keeping more than 2 providers ready rather than the bare minimum, so a single quota wall doesn't stall your benchmark run.

Your report must include:

1. **Setup**: Which models/providers were compared, which tasks were used, and why those tasks were selected

2. **Results table**: For each model × task combination:
   - Pass/Fail
   - Iterations used
   - Total input tokens
   - Total output tokens
   - Wall-clock time

3. **Provider reliability**: For each model/provider:
   - Average response time per request
   - Number of retries needed (rate limits, timeouts, errors)
   - Overall availability during benchmark runs

4. **Intermediary metrics** (at least 2 of the following):
   - Step at which the agent first reads/edits the file that appears in the final patch (exploration efficiency)
   - Step at which test failures first decrease vs baseline (partial progress)
   - Iterations between "tests first pass" and `final_answer` (submission discipline — zero is ideal)

   These metrics can be measured manually by inspecting your `solution.json` files — automation is not required. What matters is the analysis, not the tooling.

5. **Ablation study**: At least one before/after comparison of a change to your agent (prompt, tools, parameters) on the same tasks with the same model

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6. **Conclusions**: Which model(s) you selected for your final pipeline and why. Which models can be disregarded and why. Based on your data.

The backing `solution.json` files must be present in your repository.

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Chapter 5

Evaluation

During evaluation, API keys and configuration are provided via a ‘.env‘ file passed as a required argument to evaluation scripts:

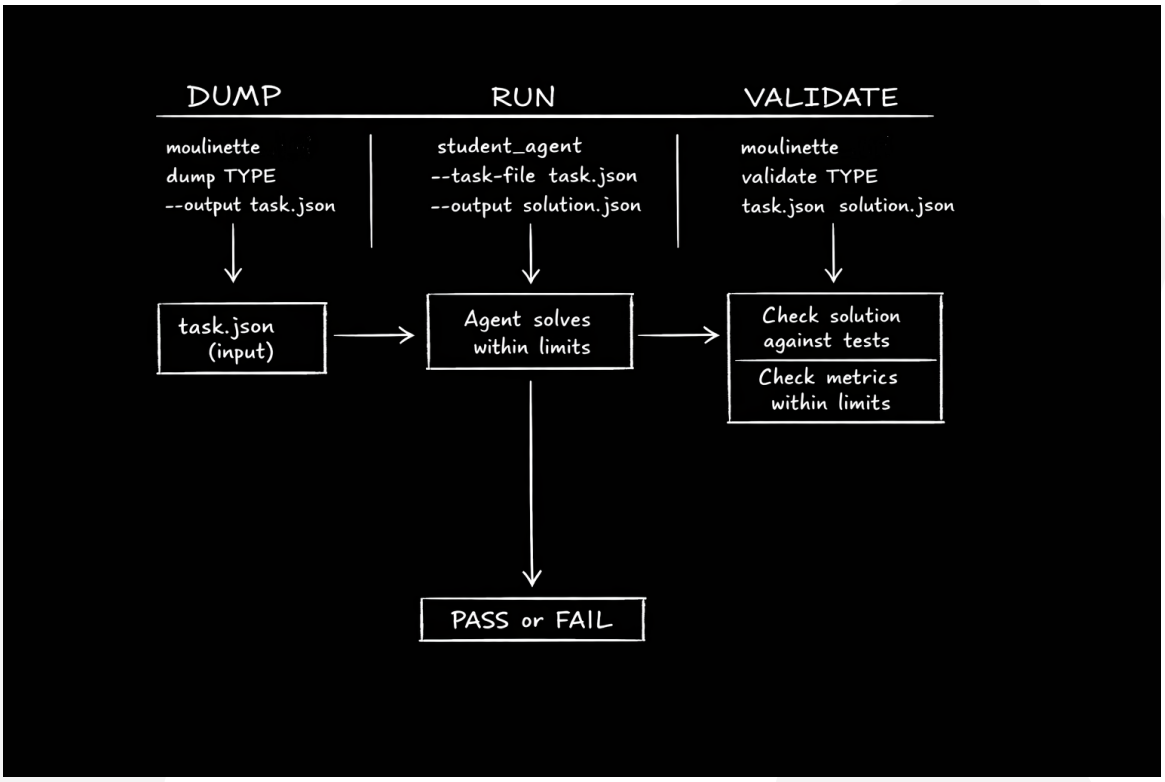
```
./exam_TYPE.sh --student-path ./student --moulinette-path ./moulinette --env-file /
   path/to/.env
```

Your CLI must support loading API keys from environment variables. Students should use standard environment variables (e.g., ‘OPENROUTER_API_KEY‘).

exam_mbpp.sh	**exam_swebench.sh**	**exam_sandbox.sh**
5 random tasks	3 random tasks	Security tests
Pass: 4/5	Pass: 2/3	Pass: ALL
For each task: 1. dump task 2. run agent 3. validate	For each task: 1. dump task 2. run agent 3. validate 4. container cleanup	Tests: - import block - builtin block - network block - path restrict - timeout - memory limit - MCP protocol

Table 5.1: Summary of exam scripts and tasks. Iteration/token/timeout limits are detailed in Section 5.1.

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5.1 Hard Requirements and Limits

Your implementation must strictly respect the following execution limits. Exceeding any of these limits will result in a failure of the corresponding task.

5.1.1 MBPP Limits

Metric	Limit
Maximum iterations	10
Maximum input tokens	6,000
Maximum output tokens	1,500
Timeout	120 seconds

5.1.2 SWE-bench Limits

Metric	Limit
Maximum iterations	30
Maximum input tokens	300,000
Maximum output tokens	10,000
Timeout	900 seconds

- Token limits are **cumulative** across all iterations of a single task (all Thought → Code → Observation cycles combined).
- Thinking/reasoning tokens (when used by reasoning models) count toward the limit just like all other tokens.
- If your chosen model's reasoning tokens make limits tight, consider using a non-reasoning model.

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- The timeout is enforced by force-killing your agent's process (and any process it spawned) with `SIGTERM` then `SIGKILL` if it hasn't returned in time, not just checked after the fact. For SWE-bench, make sure your Docker container cleanup can still run in that case (e.g. a signal handler), or complete quickly enough that it usually isn't reached.

5.2 Pass Criteria

Benchmark	Tasks	Pass Threshold
MBPP	5 random tasks	4 out of 5
SWE-bench	3 random tasks	2 out of 3

No retries are allowed during examination: a failed task is not re-run. This does not restrict your own agent's internal LLM-call retry logic, tracked by `StepMetrics.retries`.

5.3 Grading Criteria

To pass the project, you must satisfy **all** of the following conditions:

- Meet the pass criteria for both MBPP and SWE-bench
- Respect all iteration, token, and timeout limits
- All mandatory tools pass independent tests
- The sandbox passes isolation and security tests



Hardcoded API keys in your code are a security failure. All API keys must be loaded from environment variables or configuration files (`.env`). Any API key found in your source code will be flagged during evaluation.

5.4 What We Will Test in the Review

- Correctness of mandatory tools
- Correct implementation of the agent reasoning loop
- Sandbox security and isolation guarantees
- Model benchmarking results and token statistics
- Code quality, robustness, and overall architecture



During the evaluation, you will be asked to make small live modifications to your agent and re-run it on an MBPP task. This tests that you understand your own codebase and that the data in `solution.json` comes from real execution, not fabricated values.
Each modification should take 2–5 minutes. If you cannot find where to make a change, it indicates you do not understand your own codebase.
You will be asked to revert all changes after the exercise (e.g., `git checkout`).

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5.4.1 AI Safety in Evaluation

Building an autonomous code agent raises important safety considerations beyond just sandbox security. During evaluation, we verify that your agent solves tasks through legitimate reasoning and code exploration — not by exploiting shortcuts.

Your agent must **not**:

- Fetch solutions from pull requests, issues, or external sources
- Use memorized patches from its training data without genuine exploration
- Bypass the sandbox security constraints
- Access resources outside the provided task context

The `SolutionOutput` includes `system_prompt`, `llm_output`, `sandbox_input`, and `sandbox_output` fields precisely so that evaluators can trace the agent's reasoning process. This transparency is a core part of responsible AI development.

Violations will result in a grade of 0.

5.5 Evaluation logging structure

Evaluation results are stored in:

```
./evaluations/EVAL_TYPE/YYYY-MM-DD_HH-MM-SS/task_id/task.json, solution.json, stdout
    .log, stderr.log
```

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Chapter 6

Readme Requirements

A README.md file must be provided at the root of your Git repository. Its purpose is to allow anyone unfamiliar with the activity (peers, staff, recruiters, etc.) to quickly understand what the activity is about, how to run it, and where to find more information on the topic.
The README.md must include at least:

- The very first line must be italicized and read: *This activity has been created as part of the 42 curriculum by <login1>[, <login2>[, <login3>[...]]].*
- A "**Description**" section that clearly presents the activity, including its goal and a brief overview.
- An "**Instructions**" section containing any relevant information about compilation, installation, and/or execution.
- A "**Resources**" section listing classic references related to the topic (documentation, articles, tutorials, etc.), as well as a description of how AI was used — specifying for which tasks and which parts of the activity.

➠ **Additional sections may be required depending on the activity** (e.g., usage examples, feature list, technical choices, etc.).

Any required additions will be explicitly listed below.
Your `README.md` must include:

- System architecture
- Agent loop explanation
- Sandbox design
- Tool implementation details
- Benchmark results and analysis



Your README must be written in English.

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Chapter 7

Submission

Submit your project in your Git repository. Your repository must contain:

- The internal architecture and directory structure of your submission are left to you.
- Configuration files for sandbox and models
- `README.md`



Do not include Docker images, large model weights, or generated outputs.

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